The empirical recurrence for OEIS A299190, recovered from its tail

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Abstract

OEIS A299190 records "Empirical recurrence of order 69" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 71$. Its last coefficient is $c_{69} = 1024 \neq 0$, so no further tail satisfies anything shorter; any order-69 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A299190 is "Number of nX4 0..1 arrays with every element equal to 0, 1, 3, 4, 6 or 7 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$5, 13, 9, 80, 220, 518, 2466, 8609, \ \ldots$$

The entry states:

> Empirical recurrence of order 69 (see link above)

As of the "Last modified" line on the live entry (Feb 04 2018 13:25:36, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 69: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 69.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 2$ on, it returns order 69 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{69} c_i \, a(n-i),$$

c_1	4	c_{19}	36927826	c_{37}	735574685	c_{55}	9337020
c_2	4	c_{20}	−8089	c_{38}	3484233058	c_{56}	18890796
c_3	48	c_{21}	−83574731	c_{39}	243788144	c_{57}	37242230
c_4	−275	c_{22}	−109482240	c_{40}	−1266997331	c_{58}	7697378
c_5	−219	c_{23}	2599845	c_{41}	−3064364710	c_{59}	−1282168
c_6	−696	c_{24}	252727859	c_{42}	−321598251	c_{60}	−11090012
c_7	7952	c_{25}	214533427	c_{43}	1332688309	c_{61}	−7231464
c_8	4757	c_{26}	42219092	c_{44}	1847484306	c_{62}	−1868720
c_9	−1847	c_{27}	−493195689	c_{45}	338166213	c_{63}	172784
c_{10}	−127933	c_{28}	−269493200	c_{46}	−891052732	c_{64}	1067488
c_{11}	−52862	c_{29}	−347320823	c_{47}	−773693182	c_{65}	263680
c_{12}	174485	c_{30}	587724357	c_{48}	−296700475	c_{66}	132800
c_{13}	1278885	c_{31}	236501861	c_{49}	385924338	c_{67}	−3072
c_{14}	316331	c_{32}	1240979298	c_{50}	230525309	c_{68}	13056
c_{15}	−2415738	c_{33}	−355987609	c_{51}	178014110	c_{69}	1024
c_{16}	−8384559	c_{34}	−312194949	c_{52}	−98198556		
c_{17}	−882515	c_{35}	−2609385924	c_{53}	−64908392		
c_{18}	17971650	c_{36}	−16123867	c_{54}	−77530218		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 71$, and it is the recurrence OEIS A299190 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{69} - c_1 x^{69-1} - \cdots - c_{69}$. Its constant term is $-c_{69} = -1024$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 69 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 69, so $\deg q = 69$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 69, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 24 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 69, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1024.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-69 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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